

Triple-Reduction Helical Gearbox Teardown and Repair Analysis

A. Job information

Customer name

Work order number

Date received

Manufacturer

Model

Serial number

Overall reduction

: 1

Stage 1 ratio

: 1

Stage 2 ratio

: 1

Stage 3 ratio

: 1

Oil capacity

Oil grade

Application / driven equipment

Service hours (if known)

Reported failure / customer complaint

B. As-received condition

☐

Photos taken as received

Notes:

☐

External damage, cracks, missing parts

Notes:

☐

Nameplate present and legible

Notes:

Shafts turn freely by hand

☐

Yes

☐

No

Notes:

☐

Noise or roughness when turned

Notes:

☐

Oil leaks at seals / split line / plugs

Notes:

☐

Breather condition

Notes:

Ram Gear Gearbox Evaluation

C. Oil condition

Oil drained - volume recovered

Oil sample ID (lab analysis)

☐ Oil sample taken for lab analysis

Appearance: ☐ Clean ☐ Dark ☐ Milky (water) ☐ Sludge ☐ Metallic particles

Magnetic drain plug debris description

D. As-found measurements before teardown

Endplay - Input shaft

Endplay - Intermediate 1 shaft

Endplay - Intermediate 2 shaft

Endplay - Output shaft

Backlash - Stage 1

Backlash - Stage 2

Backlash - Stage 3

Ext. runout - Input shaft

Ext. runout - Output shaft

E. Teardown checklist

- ☐ Match-mark housing, caps, and shafts before disassembly
- ☐ Parts tagged and kept in order
- ☐ Seals removed and inspected

AS-FOUND SHIM RECORD

Shaft	Drive end (DE) as-found shim (in/mm)	Non-drive end (NDE) as-found shim (in/mm)
Input		
Intermediate 1		
Intermediate 2		
Output		

Ram Gear Gearbox Evaluation

F. Component inventory and repair assessment

Input shaft

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

Input pinion (stage 1)

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Teeth:

Findings:

Intermediate 1 shaft

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

Intermediate 1 gear (stage 1)

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Teeth:

Findings:

Intermediate 1 pinion (stage 2)

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Teeth:

Findings:

Intermediate 2 shaft

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

Intermediate 2 gear (stage 2)

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Teeth:

Findings:

Intermediate 2 pinion (stage 3)

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Teeth:

Findings:

Output shaft

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

Ram Gear Gearbox Evaluation

Output gear (stage 3)

☐ Reuse ☐ Repair ☐ Replace

Part no.: Teeth: Findings:

Bearing - Input DE

☐ Reuse ☐ Repair ☐ Replace

Cone P/N: Cup P/N: Findings:

Bearing - Input NDE

☐ Reuse ☐ Repair ☐ Replace

Cone P/N: Cup P/N: Findings:

Bearing - Intermediate 1 DE

☐ Reuse ☐ Repair ☐ Replace

Cone P/N: Cup P/N: Findings:

Bearing - Intermediate 1 NDE

☐ Reuse ☐ Repair ☐ Replace

Cone P/N: Cup P/N: Findings:

Bearing - Intermediate 2 DE

☐ Reuse ☐ Repair ☐ Replace

Cone P/N: Cup P/N: Findings:

Bearing - Intermediate 2 NDE

☐ Reuse ☐ Repair ☐ Replace

Cone P/N: Cup P/N: Findings:

Bearing - Output DE

☐ Reuse ☐ Repair ☐ Replace

Cone P/N: Cup P/N: Findings:

Bearing - Output NDE

☐ Reuse ☐ Repair ☐ Replace

Cone P/N: Cup P/N: Findings:

Ram Gear Gearbox Evaluation

Input seal

☐

Reuse

☐

Repair

☐

Replace

Part no.:

Findings:

Output seal

☐

Reuse

☐

Repair

☐

Replace

Part no.:

Findings:

Keys and keyways

☐

Reuse

☐

Repair

☐

Replace

Part no.:

Findings:

Spacers and locknuts

☐

Reuse

☐

Repair

☐

Replace

Part no.:

Findings:

Bearing caps and shims

☐

Reuse

☐

Repair

☐

Replace

Part no.:

Findings:

Housing and bearing bores

☐

Reuse

☐

Repair

☐

Replace

Part no.:

Findings:

Breather, plugs, sight glass

☐

Reuse

☐

Repair

☐

Replace

Part no.:

Findings:

Other:

☐

Reuse

☐

Repair

☐

Replace

Part no.:

Findings:

Other:

☐

Reuse

☐

Repair

☐

Replace

Part no.:

Findings:

Ram Gear Gearbox Evaluation

Other:

☐

Reuse

☐

Repair

☐

Replace

Part no.:

Findings:

G. Failure modes observed

Gears

☐

Pitting

☐

Spalling

☐

Scuffing/scoring

☐

Wear

☐

Tooth breakage

☐

Plastic deformation

☐

Corrosion

☐

Misaligned contact pattern

Bearings

☐

Spalling/fatigue

☐

Brinelling

☐

Contamination dents

☐

Overheating/discoloration

☐

Corrosion/etching

☐

Electrical fluting

☐

Cage damage

☐

Skidding

Location / notes

H. Dimensional checks

BEARING BORE AND SHAFT JOURNAL DIAMETERS

Shaft	Loc.	Bearing bore measured	Bearing bore spec	Bore OK	Shaft journal measured	Shaft journal spec	Journal OK
Input	DE			☐			☐
Input	NDE			☐			☐
Intermediate 1	DE			☐			☐
Intermediate 1	NDE			☐			☐
Intermediate 2	DE			☐			☐
Intermediate 2	NDE			☐			☐
Output	DE			☐			☐
Output	NDE			☐			☐

Obtain all dimensional, endplay, backlash, and clearance specs from OEM drawings and bearing manufacturer data.

Ram Gear Gearbox Evaluation

Seal journal condition notes

I. Root cause

- ☐ Lubrication failure ☐ Contamination ☐ Overload / shock load ☐ Misalignment
- ☐ Improper installation ☐ Fatigue / end of life ☐ Manufacturing defect
- ☐ Other

Root cause narrative

J. Repair summary

Parts to replace

Machining / repair work required

Recommended upgrades or changes

Estimated labor hours

Estimated parts cost

Quote number

Recommendation: ☐ Repair ☐ Replace unit ☐ Not economical to repair

K. Sign-off

Inspected by - name	Signature	Date
Reviewed / approved by - name	Signature	Date
☐ Customer approval received	Date:	